KOTLIN PROJECT

TRAVEL BOOKING SYSTEM

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KOTLIN INHERITANCE

- Inheritance is one of the key features of object-oriented programming.
- It allows user to create a newclass from an existing class.
- The derived class inherits all the features from the baseclass and additional features of its own.

ADVANTAGES AND DISADVANTAGES

» **Easy to Understand:**
Simple structure with separate parts (name, seat, etc.).

» **Reusability:**
TicketBooking class reused across different classes.

» **Clear Output:**
Labeled and easy-to-read information

» **Predefined Values:**
The details like name and seat number are fixed in the program. It would be better to allow user input to make the program more flexible.

» **No Error Checking:**
The program doesn't check for wrong or invalid inputs, which might cause issues in a real-world system.

STEPS INVOLVED

Step 1:

- Create Base Class
- Define a base class TicketBooking.
- Establish basic properties like passenger name, seat number, and travel details.

Step 2: Set Up Inheritance

- Create derived classes that inherit from TicketBooking.
- Add functions to display passenger details (name, seat number, etc.).

Step 3: Object Creation.

Step 4: Implement Methods

- Implement and call methods to print information:
- passengerName.Name()
- seatNo.SeatNo()
- arrivalLocation.ArrivalLocation()
- departureLocation.DepartureLocation()
- arrivalTime.ArrivalTime()
- departureTime.DepartureTime()

Step 5: Final Output

- Display message: “Ticket Booked Successfully”

```kotlin
open class TicketBooking(val name: String) {
    fun book() {
        println("Travels: $name")
    }
}

class PassengerName(name: String) : TicketBooking(name) {
    fun Name() {
        println("Passenger Name: $name")
    }
}

class SeatNo(name: String, val seatNo: Int) : TicketBooking(name) {
    fun SeatNo() {
        println("Seat Number: $seatNo")
    }
}

class ArrivalLocation(name: String, val location: String) : TicketBooking(name) {
    fun ArrivalLocation() {
        println("Arrival Location: $location")
    }
}

class DepartureLocation(name: String, val location: String) : TicketBooking(name) {
    fun DepartureLocation() {
        println("Departure Location: $location")
    }
}
```

```kotlin
class ArrivalTime(name: String, val time: String) : TicketBooking(name) {
    fun ArrivalTime() {
        println("Arrival Time: $time")
    }
}

class DepartureTime(name: String, val time: String) : TicketBooking(name) {
    fun DepartureTime() {
        println("Departure Time: $time")
    }
}


fun main() {
    val passengerName = PassengerName( name: "Ranjith")
    val seatNo = SeatNo( name: "Ranjith",  seatNo: 25)
    val arrivalLocation = ArrivalLocation( name: "Ranjith",  location: "Namakkal")
    val departureLocation = DepartureLocation( name: "Ranjith",  location: "Chennai")
    val arrivalTime = ArrivalTime( name: "Ranjith",  time: "10:00 AM")
    val departureTime = DepartureTime( name: "Ranjith",  time: "7:00 AM")

    passengerName.Name()
    seatNo.SeatNo()
    arrivalLocation.ArrivalLocation()
    departureLocation.DepartureLocation()
    arrivalTime.ArrivalTime()
    departureTime.DepartureTime()
    println("TICKET BOOKED")
}
```

```
C:\Users\nagal\.jdks\openjdk-22.0.2\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2024.2.1\lib\idea_rt.jar=50402:C:\Program Files\JetB
Passenger Name: Ranjith
Seat Number: 25
Arrival Location: Namakkal
Departure Location: Chennai
Arrival Time: 10:00 AM
Departure Time: 7:00 AM
TICKET BOOKED

Process finished with exit code 0
```

In this project, we created a Kotlin-based booking system using inheritance to streamline the code and improve efficiency.

THANK YOU